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2. Please provide a description of the symbol '†' used in Algorithm 9.

INTRODUCTION

TORCHes (toxoplasmosis, rubella, cytomegalovirus, herpes, and other diseases) infections are a group of congenital fetal infections¹ producing diverse neonatal sequelae with severe defects in various organs (heart and brain) to long-term neurodevelopmental delay.

It poses a diagnostic and therapeutic challenge to the obstetrician; therefore, knowledge of the interpretation of various serological tests, awareness of serious fetal consequences, and prevention is of paramount importance to guide pregnancies affected with TORCH.

The term TORCHes infection stands for (Table 1).

TABLE 1: TORCH infection complex and causative organism.

Infection	Causative organism	Type of organism
Toxoplasmosis	Toxoplasma gondii	Protozoan parasite
Others (syphilis, chicken pox, and parvovirus)	Treponema	Treponema—spirochaete
	Pallidum and varicella zoster	Bacterium
	Parvovirus B19	Varicella zoster is a member of the Herpes virus (dsDNA virus)
		Parvovirus-ssDNA virus belonging to the Parvoviridae family
Rubella	Rubella virus	*ssRNA virus
Cytomegalovirus	Cytomegalovirus	†dsDNA virus
Herpes simplex (1 and 2)	Herpes simplex virus	DNA virus

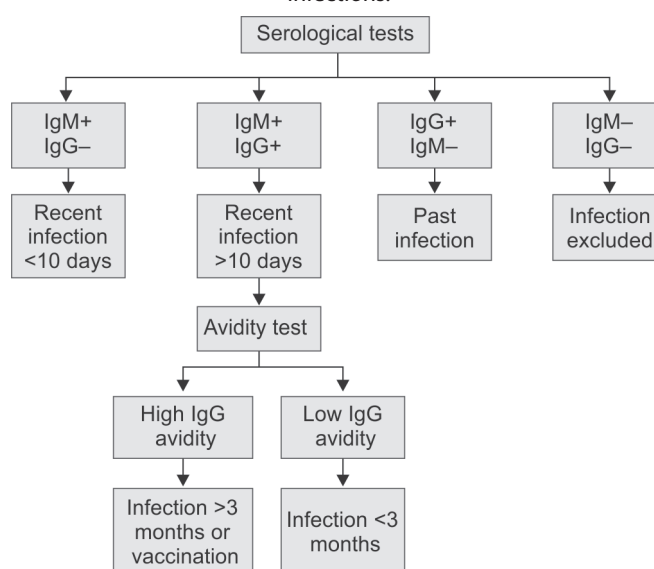
*ss: single stranded.

†ds: double stranded.

GENERAL PRINCIPLES OF TORCHES INFECTIONS SEROLOGICAL TESTS (ALGORITHM 1)

- The most commonly employed test for the diagnosis of TORCHes infection is ELISA.
- Paired serological testing for virus-specific immunoglobulin G (IgG) and immunoglobulin M (IgM) is helpful in determining the timing of infection with respect to the period of gestation.
- After the primary infection, IgM antibodies are produced within a week, but they are unable to cross the placenta due to their large molecular weight.

Algorithm 1: General principles of serological tests in TORCH infections.



- IgG antibodies are produced due to reinfection as a secondary response after a week, and they cross the placenta owing to their small size, conferring lifelong immunity.
- IgG antibody undergoes confirmation change after 3 months, allowing for antigen antibody avidity, which increases with time and is called maturation of the immune response, replacing low avidity with high avidity.²
- In case of diagnostic dilemma (acute or recent/prior infection), antibody avidity testing is advisable as it helps to determine the time elapsed since initial infection.³

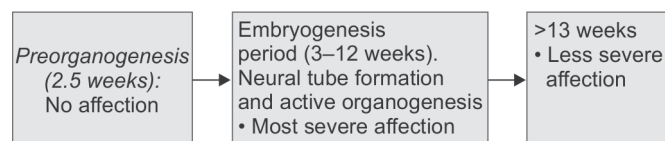
■ TIMING OF MATERNAL EXPOSURE

- The timing of maternal infection is one of the critical factors that determine whether the fetus will be infected or not.
 - Maternal infection prior to pregnancy protects the fetus by preexisting maternal IgG antibodies that have crossed transplacentally; therefore, fetal infection is very rare.
 - Acute primary maternal infection during the pregnancy can result in severe fetal outcome as the fetus is not protected by any prior IgG antibodies.

Since TORCHes infection occurs once in a lifetime, there is no subsequent threat to further pregnancy as the female is already immune due to prior infection, excluding Syphilis and very rarely infection with another strain of CMV or *Toxoplasma*. *Therefore, testing for TORCH infection is not recommended in cases of recurrent pregnancy loss.*

- The timing of maternal infection during pregnancy determines the severity of fetal infection, as the fetus is most vulnerable from week 3 to 8–12 weeks postfertilization to exposure to the pathogen (**Algorithm 2**).⁴

Algorithm 2: Depicting the relationship between the severity of fetal affection and the timing of maternal infection.



■ TESTING OF AMNIOTIC FLUID

Maternal serology only confirms maternal infection and is not indicative of fetal infection.

For confirmation of fetal infection, testing of amniotic fluid by polymerase chain reaction (PCR) is indicated, which is technically possible only after 18–20 weeks of gestation due to the following reasons:⁵

- Amniotic fluid PCR will only give positive results after 6–8 weeks of initial maternal infection.
- Fetal urination is fully established by at least 18–20 weeks, so amniotic fluid PCR will give negative results as the virus will not be detectable in sufficient quantities prior to this gestation.

This provides a very narrow window of opportunity to perform amniocentesis and taking decision based on further results, as medical termination of pregnancy (MTP) is legally not performed beyond 20 weeks of gestation (beyond 24 weeks in cases of lethal fetal malformations).

A positive PCR result confirms fetal infection, but not necessarily fetal disease, and a negative PCR does not exclude infection as sensitivity never approaches 100%.

■ INDICATIONS OF TORCH INFECTION TESTS IN PREGNANCY

- Pregnant females who present with nonvesicular rash or signs and symptoms of generalized viremia should be offered testing for rubella/parvovirus B19 (typical slapped cheek appearance).
- Ultrasonography (USG) signs shown in **Table 2** are an indication for testing for TORCH infections, although any of these signs is not diagnostic.⁶

It is the intrauterine transmission of these infections to the fetus that produces varying degrees of symptoms when the child is born. The timing of maternal infection is a key epidemiologic factor because fetal damage usually depends on the gestational age. With the exception of HSV, infections during the first trimester usually have the worst outcome.

■ TOXOPLASMOSIS

In the reproductive age group, its prevalence varies from 10% to 55% in temperate areas, which increases up to 80% in tropical areas, and is usually self-limiting with

TABLE 2: USG signs presumptive of TORCH infections.

Cranial abnormalities	Extracranial abnormalities	Placental/amniotic fluid abnormalities
• Ventriculomegaly • Calcifications • Intraventricular synechiae • Cerebellar abnormalities (vermian hypoplasia, Cerebellar hemorrhage, Calcifications, cysts) • Periventricular pseudocysts • Malformations of cortical development (lissencephaly, Polymicrogyria, schizencephaly)	• Small for gestational age • Hyperechogenic bowel • Liver calcifications • Ascites • Pericardial effusion • Subcutaneous edema • Hydrops or fetal anemia (MCA-PSV >1.5 MoM) in absence of maternal atypical antibodies	• Placentomegaly • Placental calcification • Oligohydramnios/anhydramnios • polyhydramnios

exposure to contaminated soil, undercooked meat, and unfiltered water.⁷

Epidemiology

The life cycle includes two hosts:

- **Definitive host:** Cats and felines.⁸
- Sexual reproduction takes place, and oocytes are formed, which are passed via a cat's fecal matter, thus contaminating soil and water.
- **Intermediate hosts:** Humans, pigs, and lambs who accidentally ingest oocytes, develop parasitemia which may last up to 10 days, and thereby can infect any organ where it gets transformed into a dominant cyst and remains for life long.

CLINICAL FEATURES OF FETOMATERNAL TOXOPLASMOSIS (ALGORITHMS 3A AND B)^{9,10}

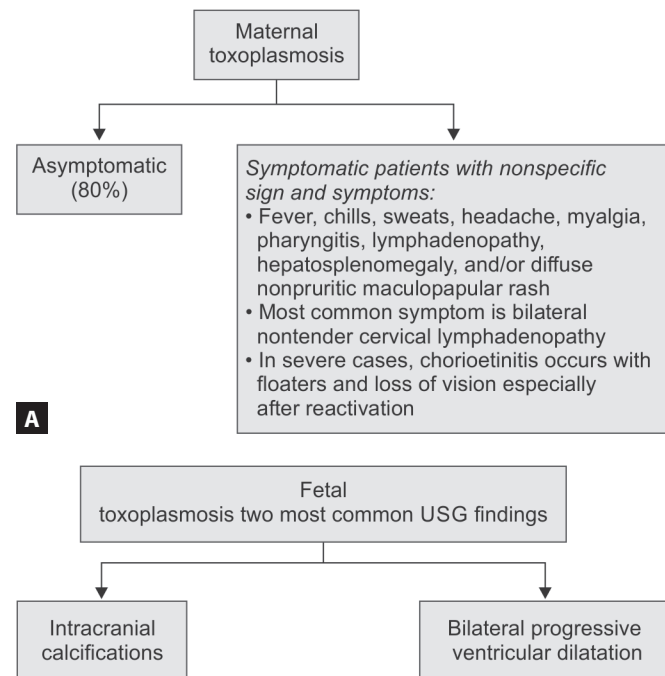
Clinical Presentation (Algorithms 3A and B)

See Algorithms 3A and B.

Neonatal Toxoplasmosis

- Asymptomatic (80%)¹¹
- Symptomatic (20%)
 - Two-thirds have moderate disease with intracranial calcifications and peripheral retinochoroiditis.¹²

Algorithms 3A and B: Clinical presentation.



- One-third have disseminated toxoplasmosis with hydrocephalus or macular chorioretinitis.

Maternal to Child Transmission

Increased risk of maternal to child transmission (MTCT) occurs due to the following factors:

- Acute primary infection during pregnancy or in the preconception period (within 3 months).
- Reactivation of infection in an immunocompromised pregnant female due to human immunodeficiency virus (HIV) infection/on immunosuppressive medications.
- Lack of antepartum treatment.
- Reinfection with a more virulent strain with high parasitic load.

No evidence of transmission via breastfeeding or direct human-to-human contact.

Sources of Infection

- Contact with contaminated soil/water, raw milk, or undercooked meat.
- Transplacental transfer of oocytes occurs, which increases with placental maturation, and therefore risk rises with an increase in gestation age (risk of MTCT is lower in early gestation).

Poor prognostic factors on USG:

- Severe ventricomegaly
- Necrosis of the brain
- Gyration disorder
- Microcephaly

Diagnosis of Fetal Infection

- Detection of **Toxoplasma deoxyribonucleic acid (DNA)** by PCR on fetal blood/amniotic fluid is the most reliable method of prenatal diagnosis with a positive predictive value (PPV) = 100% and specificity-100.
- A negative PCR cannot completely rule out fetal infection; therefore, continued follow-up is advisable by a serial USG, prophylaxis with spiramycin therapy, and full neonatal examination.
- *Amniocentesis is indicated in the following cases:*
 - Primary maternal acute infection
 - Serological testing cannot confirm or exclude infection
 - Presence of abnormal cranial findings, ascites, hepatosplenomegaly, and severe fetal growth retardation.

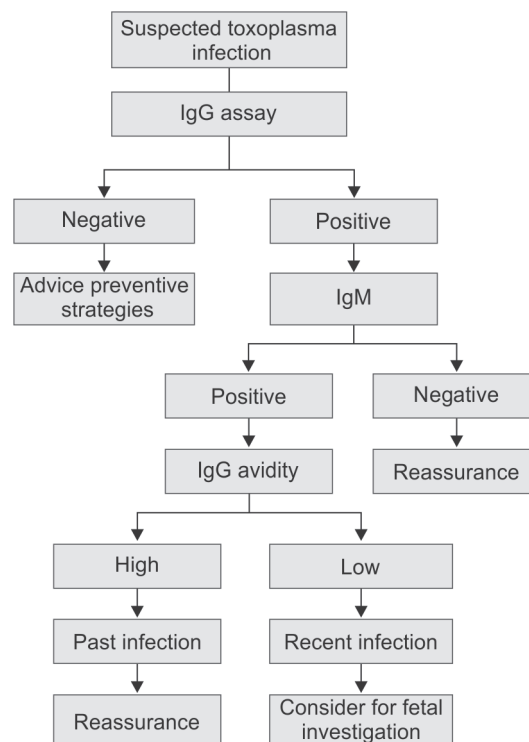
Amniocentesis is not recommended in pregnant female who are HIV/HBsAg positive due to procedural related risk of fetal transmission.

Diagnosis of Toxoplasmosis

- The goal of the investigation is to establish the diagnosis in the mother while detecting the risk and severity of fetal and neonatal infection (**Algorithm 4**).
- All guidelines do not recommend universal screening of Toxoplasma, and maternal screening is only indicated in the presence of abnormal USG findings.
- Low avidity is not always diagnostic of recent infection, as it can persist for years in some females.
- If both IgG and IgM are positive, a second sample should be sent to the laboratory for further confirmation.¹³

Diagnosis in Newborns

- Newborns should be thoroughly examined by a multidisciplinary team that comprises an expert neonatologist, retinal specialist, and pediatric neurologist.
- IgM, IgG, and IgA serological testing and PCR of blood, urine, and cerebrospinal fluid (CSF) of the neonate should be done as early as possible, soon after birth.¹⁴

Algorithm 4: Diagnosis of maternal toxoplasmosis.

(IgG: immunoglobulin G; IgM: immunoglobulin M)

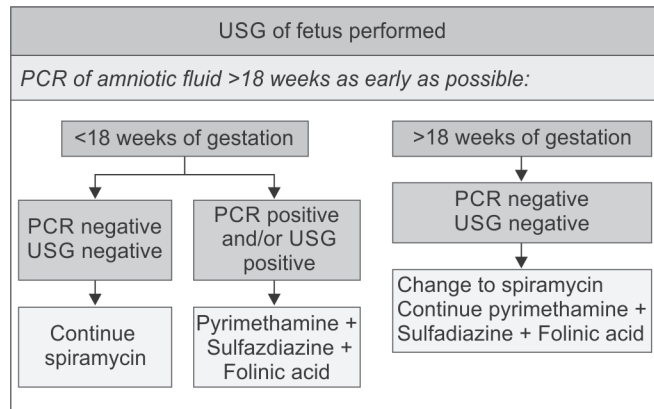
TABLE 3: Treatment of toxoplasmosis.¹⁵

Treatment of toxoplasmosis in pregnancy	
<18 weeks of gestation	≥18 weeks of gestation
Spiramycin: 1 g TDS9	Pyrimethamine 50 mg/day + Sulfadiazine, three tablets per day of 500 mg + Folinic acid two capsules of 25 mg/week 2 L of water/day is advisable to alkalinize urine
Contraindications (C/I): Long QT syndrome	C/I: G-6PD deficiency
Side effects (S/E): GIT symptoms (nausea, vomiting), and skin reactions	S/E: Skin reaction, bone marrow suppression, and reversible acute kidney injury
No monitoring required	Monitoring: CBC every week

Management of Toxoplasmosis (Table 3) (Algorithm 5)

- General routine prenatal care has to be followed for pregnant patients who are diagnosed with active toxoplasmosis infections.

Algorithm 5: Treatment according to diagnosis of fetal Toxoplasmosis.



(PCR: polymerase chain reaction; USG: ultrasonography)

- Following fetal infection diagnosed by PCR, follow-up with monthly fetal USG is recommended for early diagnosis of markers suggestive of affection.
- Should be managed in coordination with a fetal medicine expert and an obstetrician.
- Congenital toxoplasmosis does not affect the decision of timing and mode of delivery.
- Breastfeeding is not contraindicated, although the mother should be cautious if cracked or bleeding from nipples are present.

Prevention

Primary Prevention

- Meat should be properly cooked at 63–74°C temperature, before its consumption.¹²
- If refrigeration is required, meat should be kept for a minimum of 72 hours.
- Drinking unpasteurized goat milk, butchering or skinning an animal without gloves, and contact of the mucous membrane and handling meat with bare hands should be avoided.
- Strict hand hygiene and safe sanitation practices should be strictly followed.
- Handling of cat feces should be avoided.

Secondary Prevention

In the presence of low prevalence and incidence of maternal infection, routine universal screening is not recommended, as it is not cost-effective.¹⁶

■ SYPHILIS

Modes of Transmission

- Direct contact with an infectious lesion.**
- Vertical transmission during pregnancy:* Risk of transmission varies with stage of disease and is greatest in early-stage disease and increased during the third trimester.
- Sexual contact:* One-third of cases are sexually transmitted.

■ WORLD HEALTH ORGANIZATION RECOMMENDATIONS FOR SCREENING OF SYPHILIS

Universal screening for syphilis is recommended for all pregnant females as soon as possible.

Objectives of Universal Screening

- To decrease the risk of vertical transmission of syphilis
- Identification of syphilis in the initial stages of pregnancy.
- Timely referral of the neonate for appropriate management.
- All pregnant females should be universally screened for syphilis during the first antenatal visit, irrespective of high or low prevalence of syphilis or any past result or in previous pregnancy.
- Single on-site rapid syphilis test (RST) is preferred over single on-site rapid plasma reagin (RPR) to screen pregnant females in low-prevalence areas (<5%).¹⁷
- In areas with high prevalence (>5%), RST is performed, and the first dose of treatment is given if found to be positive, followed by the RPR test. Further treatment is administered according to the duration of syphilis.
- A repeat test should only be done if a woman has been at risk of reinfection after initial screening or has signs and symptoms of infection, such as symptoms of STI, and a patient who is HIV positive.
- Pregnant female with positive screening should be considered infectious unless the history of treatment is elicited.

■ CLINICAL MANIFESTATIONS

Syphilis is a multistage and multisystemic disease that can be either congenital or acquired (**Table 4**).

TABLE 4: Clinical features of acquired syphilis.

Early	IP-9-90 Days (21 days) <i>Clinical presentation (C/P):</i> Single painless papule with regional lymphadenopathy, which ulcerates followed by Chancre formation (classically)
Latent/ secondary	Occurs after months in 25% of patients <i>C/P:</i> Widespread mucocutaneous papular rash with generalized lymphadenopathy
Tertiary	Occurs approximately after 20–40 years in one-third of patients: • Gummatous • Neurosyphilis • Cardiovascular syphilis

ADVERSE PREGNANCY OUTCOMES

- Fetal anemia
- Fetal growth restriction (FGR)
- Hepatosplenomegaly
- Intrahepatic calcification
- Fetal hydrops
- Stillbirth

MANAGEMENT

When not to treat:

- Management in case of complete treatment of syphilis prior to the current pregnancy.
 - No treatment is required for mother as well as neonate if the RPR is negative at the first antenatal visit as well as in repeated testing in case of risk of reinfection.
- No need for maternal treatment in cases of:
 - Biological false positive test.¹⁸
 - History of adequate treatment before index pregnancy

When to treat:

- Retreatment in a female who was treated before conception is considered only if:
 - History is suggestive of inadequate treatment.
 - Inadequate serological cure [fourfold drop in RPR/venereal disease research laboratory (VDRL) titer]
- Management in cases of positive treponemal serology:¹⁹
 - In females where treatment history could not be properly obtained, syphilis is staged as early when there is the presence of a clinical lesion and/or positive PCR or presence of *Treponema pallidum* on dark ground microscopy.

Till 27 ₊₆ weeks	Benzathine penicillin G 2.4 MU intramuscular single dose
28 weeks till term	Benzathine penicillin G 2.4 MU intramuscular on day 1 and 8 with careful assesment of neonate at birth

Fig. 1: Treatment of early syphilis.

- Disease is presumed to be late stage if negative results are unavailable.
- Maternal treatment is indicated in cases of:
 - Any stage of active syphilis.
 - Doubtful history of treatment prior to index pregnancy.

TREATMENT OF SYPHILIS IN PREGNANCY¹⁹

The aim of the treatment of syphilis during pregnancy should be:

- Complete eradication of disease and decrease risk of reinfection.
- Treatment of fetal infection if it occurs at all.

Early Syphilis (Fig. 1)

- WHO STI guideline recommends benzathine penicillin G 2.4 million units once intramuscularly over procaine penicillin in pregnant women with early syphilis.
- In cases of penicillin allergy, desensitization must be tried, as parenteral penicillin is the most effective treatment available during pregnancy.
- If desensitization is not possible, erythromycin 500 mg orally four times daily for 14 days or ceftriaxone 1 g intramuscularly once daily for 10–14 days or azithromycin 2 g once orally can be given as second-line treatment, but they are associated with treatment failure and adverse pregnancy outcome.
- Babies born to mothers who were treated with macrolide or inadequate treatment should be treated for congenital syphilis.
- Treatment of the partner is critical to prevent reinfection.
- Serological tests may be done at delivery to identify reinfection, which helps in the diagnosis of congenital syphilis.

Late Syphilis/Cardiovascular/Gummatous Syphilis

Benzathine Penicillin G 2.4 MU intramuscular weekly on days 1, 8, and 15 (three doses) is preferred over procaine penicillin 1.2 million units intramuscularly once a day for 20 days.

In case of penicillin allergy, if desensitization is not possible or penicillin is not available, erythromycin 500 mg orally four times daily for 30 days can be given.

Neurosyphilis¹⁹

Procaine penicillin G 1.8–2.4 MU intramuscular OD, along with probenecid 500 mg per oral four times a day for 14 days.

Or

Benzyl penicillin 10.8–14.4 g daily, given 1.8–2.4 g intravenously every 4 hours for 14 days.

Note: Steroids should be given with all antitreponemal antibodies for neurosyphilis and cardiovascular syphilis in a dose of 40–60 mg prednisolone OD for 3 days, which should be started 24 hours prior to antibiotics.

Congenital Syphilis

- May present early (<2 years of life) and late (>2 years of life).
- Occurs via transplacental transmission or during delivery in the presence of maternal genital infection

- Neonate should be treated for congenital syphilis if therapy has been administered in less than 4 weeks prior to delivery.

Diagnosis of Congenital Syphilis

- All infants born to mothers with positive serology require detailed clinical examination and serological testing unless positive serology was confirmed to be a false positive or when syphilis was completely treated previously (Algorithms 6 and 7).

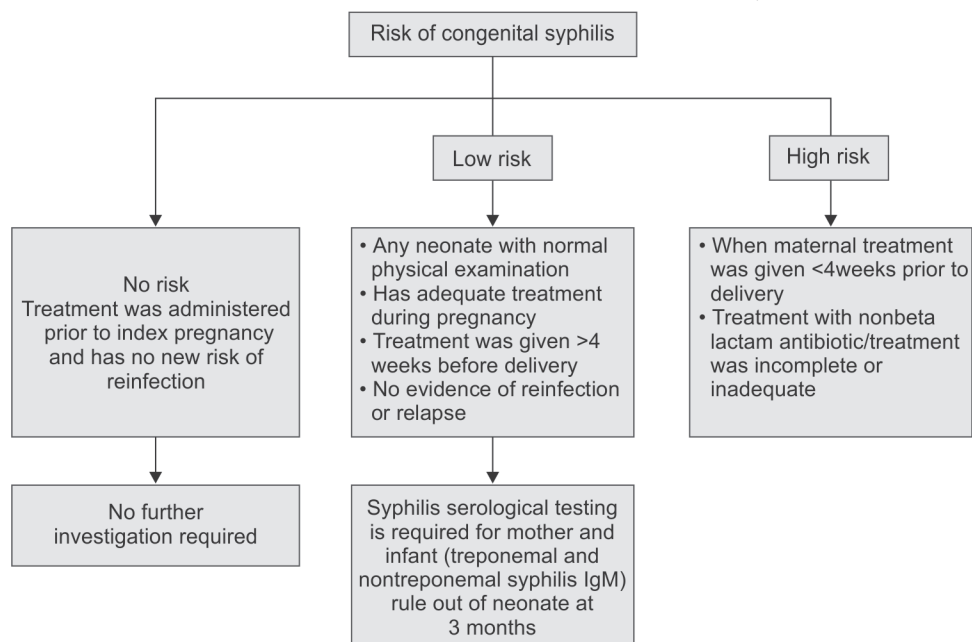
RUBELLA

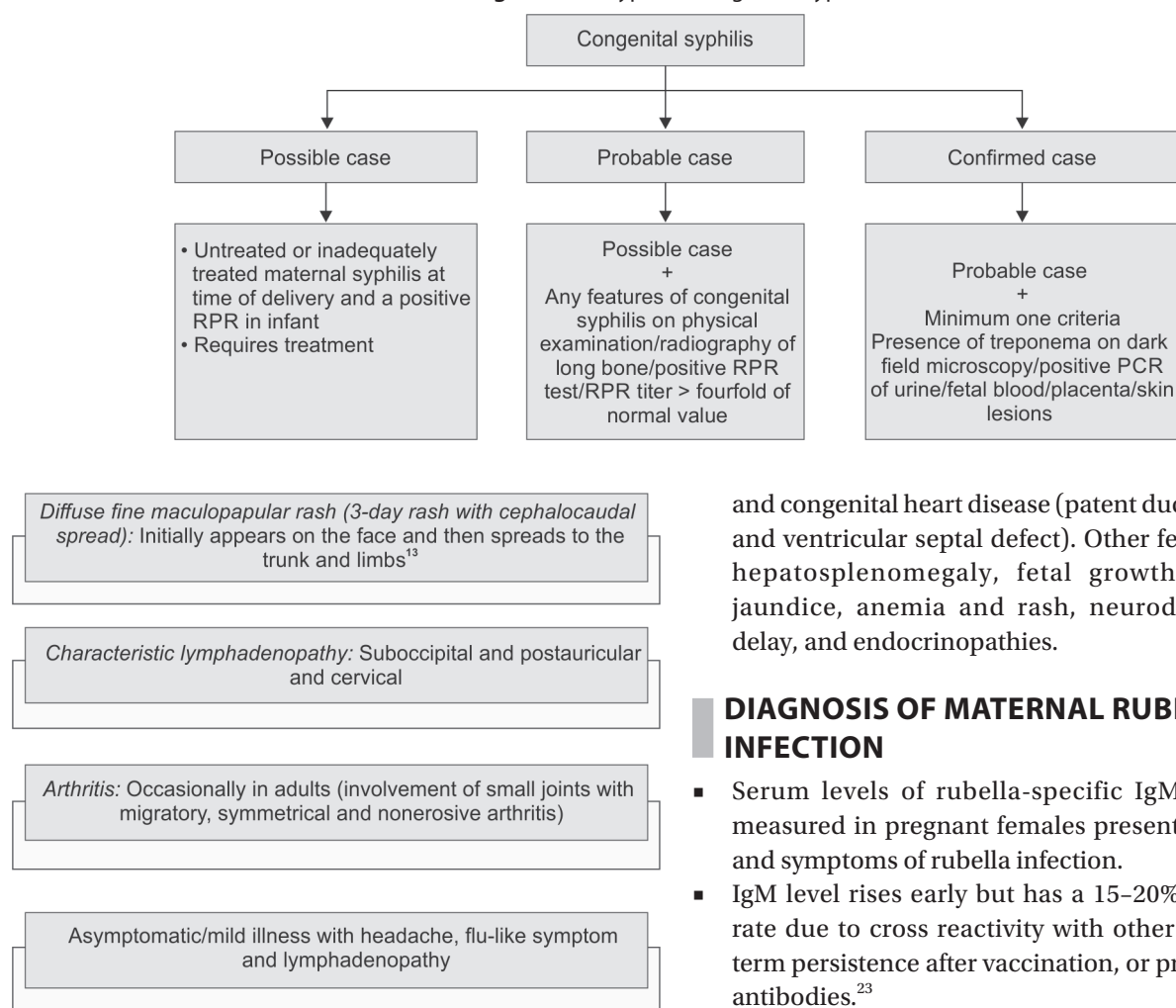
- Rubella is a respiratory tract illness caused by the rubella virus belonging to the Togaviridae family with an incubation period of 14 days (12–23 days).
- Wide-spread rubella vaccination has led to high levels of herd immunity within the community (98–99%), therefore routine testing for rubella during pregnancy is not recommended nowadays.²⁰

CLINICAL MANIFESTATIONS IN ADULTS (FIG. 2)

The infectivity period begins 7 days before the appearance of rash and persists for 7–10 days after its onset. Patients may be asymptomatic or may be present with severe symptoms.

Algorithm 6: Approach according to the risk of congenital syphilis.



Algorithm 7: Types of congenital syphilis.**Fig. 2:** Clinical manifestations of rubella.

CONGENITAL RUBELLA SYNDROME

- Risk of fetal transmission decreases with advancing gestational age at maternal infection.
 - <12 weeks: 90%
 - 12–16 weeks: 55%
 - >16 weeks: 45%
- Risk of developing congenital defects is greatest when infection occurs before 12 weeks, that is 97%, and decreases to 20% between 12 and 16 weeks. Beyond 16 weeks, rubella infection may be associated with minimal risk of deafness only.^{21–23}
- Clinical consequences of chronic rhinosinusitis (CRS) include a triad of sensorineural hearing loss,¹⁴ cataracts,

and congenital heart disease (patent ductus arteriosus, and ventricular septal defect). Other features include hepatosplenomegaly, fetal growth retardation, jaundice, anemia and rash, neurodevelopmental delay, and endocrinopathies.

DIAGNOSIS OF MATERNAL RUBELLA INFECTION

- Serum levels of rubella-specific IgM and IgG are measured in pregnant females presenting with signs and symptoms of rubella infection.
- IgM level rises early but has a 15–20% false positive rate due to cross reactivity with other viruses, long-term persistence after vaccination, or presence of auto antibodies.²³
- Diagnosis of acute infection should be based on history of exposure/vaccination/development of rash and the result of previous rubella tests. IgG avidity helps to differentiate between recent and old infection and should be done when both IgG and IgM are positive.²⁴
- Viral ribonucleic acid (RNA) detection via PCR from pharyngeal exudates can also confirm the primary maternal infection.

DIAGNOSIS OF FETAL RUBELLA INFECTION

- Termination of pregnancy should be considered even without invasive testing if primary infection occurs before 12 weeks, considering the increased risk as well as severity of fetal infection.²⁴
- Invasive testing (Amniocentesis) for detection of viral RNA is advisable if primary maternal infection

occurs between 12 and 20 weeks, as the risk of fetal transmission is less, or the presence of ultrasound markers in rubella infection.

- Amniocentesis, if performed within 6 weeks of primary infection, is more likely to give a false negative result; therefore, repeat invasive testing is justifiable in some conditions.

■ VACCINATION

- Measles, mumps, and rubella vaccination (MMR) is a live-attenuated vaccine and therefore is contraindicated during pregnancy.
- It is recommended to avoid pregnancy for 1 month after administration of the rubella vaccine.
- Breastfeeding does not contraindicate vaccination.
- Rubella immunity should be ensured in all women prior to pregnancy.

■ CYTOMEGALOVIRUS

- Cytomegalovirus (CMV) is the most common perinatal viral infection, affecting 0.2–2.2% of all live births.^{25–28}
- Most common cause of nongenetic sensory neural hearing loss.²⁷
- Major cause of neurological disabilities
- It remains dormant lifelong, especially in salivary glands, and can be reactivated anytime; therefore, the most common source of infection is saliva/urine in young children.

Clinical Features of Maternal Cytomegalovirus Infection

- *Asymptomatic*: 85–90%²⁹
- *Symptomatic*: 10–15%. Clinical features similar to mononucleosis include fever, malaise, cervical lymphadenopathy, and sometimes hepatitis.
- Infection may be acquired for the first time during pregnancy or result from reactivation of prior infection or reactivation of infection with different strains (nonprimary infection).
- Routine antenatal CMV screening is not recommended as it hardly meets any screening criteria, as well as the availability of effective treatment.³⁰

Risk of Transmission of Fetal Infection

- Risk of transmission increases to 30–40% after primary CMV infection as compared to 1–2% after nonprimary maternal CMV infection.²⁸

- Risk of transmission increases to approximately 30% in early pregnancy as compared to around 47% when infection occurs in the third trimester.³¹
- Risk of severity of fetal infection is higher when infection occurs in the first trimester as compared to the late gestational period.
- The most severe damage to neurological and hearing organ systems occurs almost exclusively when the fetus becomes infected during the first trimester of pregnancy.³²
- *Clinical features in congenital CMV infection at birth*: Small for gestational age, microcephaly, jaundice, hepatosplenomegaly, hypotonia, blue muffin rash, chorioretinitis, and visual deficit.
- Sensory neural hearing loss and mild to moderate neurodevelopmental sequelae may present at a later stage, which underscores the importance of long-term follow-up of such babies by the pediatrician.

Diagnosis of Maternal Cytomegalovirus Infection

Indications of serological testing for CMV in mother:^{33,34}

- Influenza-like symptoms or symptoms of glandular fever with negative results for the Epstein-Barr virus.
- Symptoms of hepatitis with negative test results for hepatitis A, B, and C during pregnancy.
- Fetal ultrasound suggestive of possible CMV infection.

Interpretations of Serological Tests

The presence of CMV-specific IgM is commonly not diagnostic of acute infection due to the following reasons:

- Persistence of IgM for a longer duration after primary CMV infection.
- Presence of IgM due to nonprimary secondary infection.
- Cross reactivity with other virus like Epstein-Barr virus.
- Nonspecific polyclonal stimulation.



IgG specific for CMV along with its avidity should be done in seropositive females to determine the timing of maternal infection (<30% avidity is suggestive of recent primary infection whereas >60% is indicative of past infection if for >3 months.)

- Diagnosis of nonprimary infection and transmission to the fetus is done by invasive testing via amniocentesis for detection of CMV by PCR, as it is the only available method to confirm infection from different strains.

Diagnosis of Fetal Cytomegalovirus Infection

- Diagnosis is made by detection of CMV DNA in amniotic fluid or cord blood, ideally after 20 weeks period of gestation or at least after 8 weeks of maternal infection.²⁴
- *Prognostic indicators in congenital CMV infection include:*
 - Timing of infection has already been discussed above
 - *Laboratory parameters:*
 - ♦ Median viral load in amniotic fluid is higher in symptomatic as compared to asymptomatic.
 - *Presence of fetal abnormalities:*
 - ♦ The most important ultrasound prognostic indicator for symptomatic fetal CMV infection is cerebral abnormality
 - ♦ Normal fetal brain findings on ultrasound and magnetic resonance imaging (MRI) are associated with minimal risk of neurodevelopmental disability, although not prognostic of hearing impairment.
 - ♦ Although fetal ultrasound abnormalities can appear as early as 12–14 weeks after maternal infection, follow-up ultrasound every 2–4 weeks should be done for the remaining duration of pregnancy.

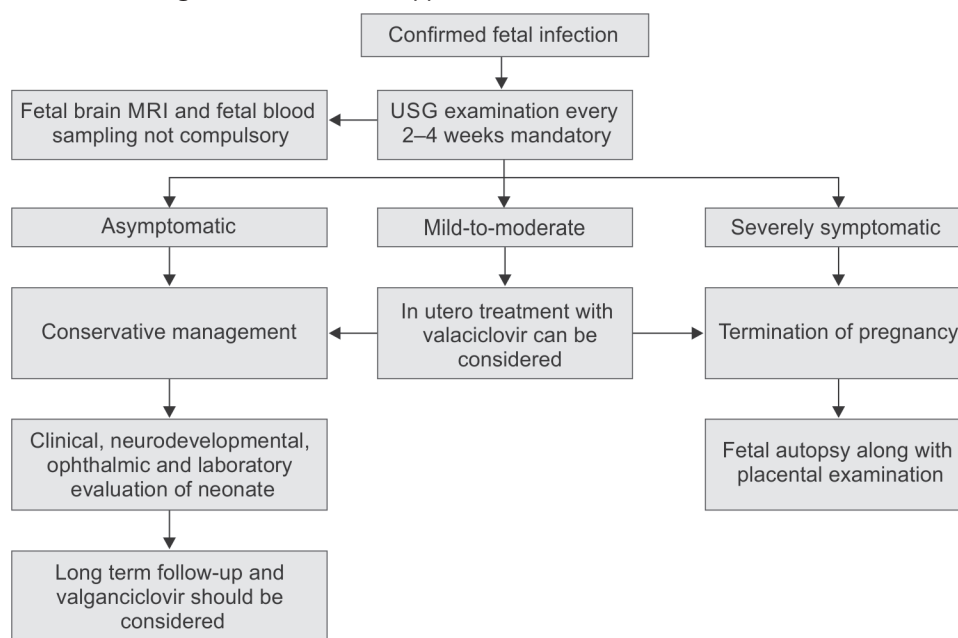
Treatment of Cytomegalovirus Infection (Algorithm 8)

- Management of CMV infection is quite challenging as diagnosis at times is difficult, and the availability of treatment options are limited.
- Treatment includes either continuation of pregnancy with regular fetal monitoring or termination of pregnancy.
- High-dose valacyclovir and CMV-specific hyper-immunoglobulin for congenital CMV infection should only be given in research trials and are not routinely prescribed.²⁴

Neonates with Symptomatic Congenital Cytomegalovirus Infection

- Congenital CMV infection should always be confirmed at birth as early as possible, even when it was previously confirmed in utero by invasive testing.
- A salivary or urine sample from the neonate should be collected within 3 weeks of birth for CMV PCR to confirm congenital rather than natal transmission.
- Valganciclovir should be given within the first 4 weeks of life³⁵ as it prevents progression of neonatal symptoms and improves long-term neonatal outcome.

Algorithm 8: Treatment approach in confirmed fetal CMV infection.



■ HERPES SIMPLEX VIRUS

Two subtypes → Herpes simplex virus (HSV) type 1 and type 2:

- 40% of the genomic structure of viruses are similar; therefore, heterologous antibodies are produced that neutralize both types of viruses.
- HSV 1 clinically causes orofacial herpes, and HSV-2 mainly causes genital herpes, but both subtypes can lead to either type of infection.
- Most common STI in the reproductive age group.
- Transmission of the virus occurs via direct and close contact.
- Seroprevalence in the reproductive age group is approximately 60–75% and 11–30% for HSV type 1 and 2, respectively.
- Incubation period is 2–12 days.

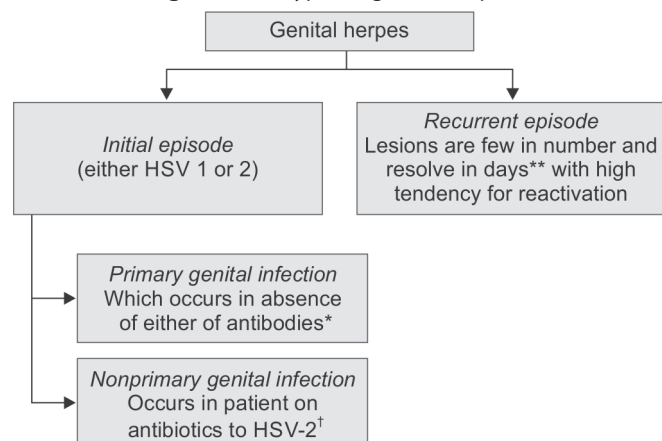
Classification

Genital herpes may present as an initial episode or a recurrent episode (**Algorithm 9**).

Factors Determining The Risk of Vertical Transmission³⁷

- Timing and type of maternal infection.
- Presence of active genital lesions.
- Presence of trans placentally acquired maternal neutralizing antibody.
- Duration of rupture of membranes prior to delivery as well as mode of delivery.¹⁷

Algorithm 9: Types of genital herpes.³⁶



*Ranges from an asymptomatic state or may be associated with severe symptoms, including herpetic lesions with inguinal lymphadenopathy

†Short duration of illness without systemic manifestation

- Use of assisted vaginal delivery (AVD). Use of fetal scalp electrode.

- Risk being greatest when there is primary initial infection in the third trimester, especially within 6 weeks of delivery, due to active viral shedding and absence of protective maternal antibody.
- As the majority of primary infections are asymptomatic, it becomes quite challenging to differentiate between a recent and a primary episode.
- Three types of perinatal transmission have been associated with HSV infection (**Algorithm 10**).

Diagnosis

- Cell culture/PCR of genital lesions.³⁸
- A sample is taken from the base of the lesion with a nylon swab and placed within a tube in transport media and stored at 4°C.
- The sensitivity of culture is 80–90% in primary active infection.
- A negative culture does not rule out genital herpes infection because the virus is shed intermittently.
- Serology has limited use as it is nonspecific and therefore is not indicated in patients with lesions suggestive of herpes, nor is it useful in pregnancies, ultrasound findings compatible with congenital infection.
- Type-specific serology is only indicated in a pregnant woman with a first episode of genital herpes in the last weeks of pregnancy (diagnosed by PCR or culture) to confirm the absence of homologous antibodies, as decision for mode of delivery has to be decided. Elective LSCS is indicated in active HSV-2 genital lesions.

Herpes simplex virus deoxyribonucleic acid in amniotic fluid (HSV-DNA) in AF has high sensitivity

Algorithm 10: Types of perinatal transmission.

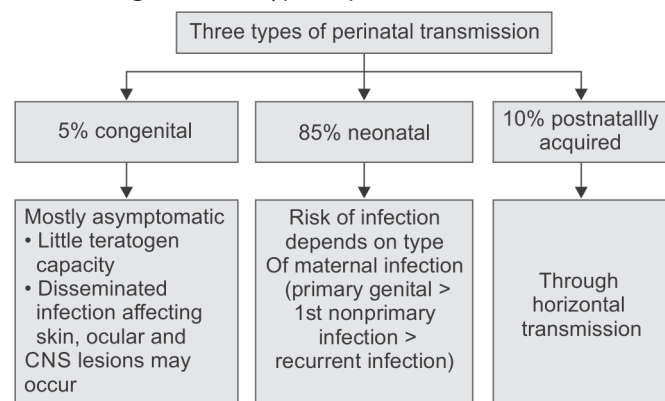


TABLE 5: Prevention of MTCT.

	<i>Primary infection</i>	<i>Recurrent infection</i>
<i>Acute episode</i> Acyclovir oral 400 mg/8 hours Valaciclovir 500 mg to 1 g/12 hours	7–10 days	3–5 days
<i>Suppressive regimen</i> Acyclovir oral 400 mg/8 hours Valaciclovir 500 mg/12 hours	>36 weeks until delivery	>36 weeks until delivery
Cesarean section	If there is an infection at the time of delivery	If an active episode at the

and specificity for the diagnosis of congenital infection by HSV-1 and 2, and is done in the presence of USG abnormalities compatible with HSV infection.

Treatment^{39,40}

- **Aim:** Prevention of mother-to-child transmission (MTCT) and treatment varies according to the type of episode, as discussed in **Table 5**.
- Other management of the symptomatic first episode may include use of supportive therapies, including paracetamol and topical lidocaine 2% gel or 5% ointment.^{36,41}
- In case of severe maternal herpes or disseminated herpes, IV acyclovir (5–10 mg/kg every 8 hours for 2–7 days) is administered, and treatment is continued with oral therapy for 10 days.
- In case of inevitable vaginal delivery or preference for vaginal delivery, invasive procedures are to be avoided, like fetal blood sampling and prolonged rupture of membranes. Here, and in case of ≥35 weeks of gestation, active management with immediate induction of labor is indicated.
- Serial cultures to identify asymptomatic HSV excretion are not indicated.

■ PARVOVIRUS B19

- It is a common childhood viral illness, also known as fifth disease or erythema infectiosum, and the only member of the family that causes human disease.
- Approximately 60–75% of pregnant females are immune to this virus.^{33,34,42}

- It often occurs during the winter and spring seasons as an epidemic in children exhibiting a characteristic facial rash known as the slapped cheek appearance.
- Transmission is via respiratory droplets from infected people by the transplacental route and rarely by blood and blood products.
- Incidence of acute infection is approximately 1–2% in pregnant females.^{34,43,43}
- Incubation period is 4–14 days, and females remain infectious for approximately 2–10 days postexposure or till the rash appears.
- *Risk of vertical transmission:* 25–32%^{35,44,45}
- The most common indication for testing of parvovirus infection is a history of recent exposure or incidental finding of nonimmune fetal hydrops.
- Although parvovirus is nonteratogenic, fetal medicine experts advise cerebral imaging in fetuses or neonates who have a severe degree of fetal hydrops, which can be an independent risk factor for neurological sequelae.
- The most important receptor is blood group antigen “P,” which is predominantly present in erythroid precursors, myocardium, and placental tissue. Therefore, it can lead to fetal anemia and aplastic crisis by inhibition of erythropoiesis.

Indications of Maternal Parvovirus Testing

- Pregnant women with a history suggestive of a peculiar rash and a hydropic fetus on ultrasound should be tested for parvovirus B19-specific IgM and IgG antibodies.
- Women with negative IgM and positive IgG are indicative of previous immunity; therefore, such females are not at risk of transplacental transmission.
- 20–40% of false negative IgM has been reported; therefore, in such suspicious cases (hydropic fetus), it should always be supplemented by the detection of viral DNA via amniocentesis as well as maternal serological IgG avidity.³⁶

Diagnosis of Fetal Parvovirus Infection

- Fetal infection can only be confirmed by amniotic fluid PCR (sensitivity 100%) obtained by invasive testing.
- Invasive testing is not usually indicated unless cordocentesis is being done for intrauterine transfusion in cases of fetal anemia.

Management²¹

- Ultrasound monitoring for evidence of hydrops, ascites, cardiomegaly, increase in MCA PSV (>1.5 MoM) should begin 4 weeks after maternal exposure and be performed every 1–2 weeks for 2–3 months after infection.
- MSA PSV >1.5 MoM predicts severe fetal anemia with a sensitivity of 95% in confirmed parvovirus B19 infection.
- MCA PSV may not detect cases of mild fetal anemia, as it does not increase until the hemoglobin is <3 g/dL and after 33 weeks period of gestation.
- MCA PSV should always be performed during periods of fetal inactivity.
- Intrauterine transfusion is most commonly in evidence of MCA PSV >1.5 MoM, evidence of fetal ascites/hydrops.

■ VARICELLA ZOSTER VIRUS

- It is a highly contagious virus and transmission occurs via:
 - Respiratory droplets
 - Direct contact with vesicular fluid
 - Indirectly via fomites
- Primary infection in pregnancy is rare (3 in 1,000) as 90% of females are immune to VZV due to previous infection usually in childhood.³⁷

Diagnosis of Maternal Varicella Zoster Virus Infection

- *Incubation period:* 7–21 days.
- Patient is infectious 48 hours prior to the appearance of rash till crusting phase of the vesicle.
- VZV during pregnancy can be associated with varicella pneumonia, which is associated with maternal morbidity and increased risk of perinatal morbidity and mortality.
- Majority of the time, diagnosis is clinical, and serological testing is usually not required if the mother has a history of clinical infection or presence of rash (rash is maculopapular in initial stages and gradually becomes vesicular associated with pruritis followed by crusting).²¹
- Nonimmune pregnant females are considered at high risk of VZV infection if exposed to an infected person

(face to face for 5 minutes or in the same room for 15 minutes).

- Risk of congenital varicella syndrome is 0.5% if the mother is exposed in the first 13 gestational weeks and 2% between 13 and 20 weeks, thereafter the risk is minimal.³⁸
- If the maternal infection occurs after 36 weeks, there is a risk of 50% fetal infection and 25% risk for neonatal varicella.
- Maternal herpes zoster (Shingles) does not possess any risk of infection for the fetus.³⁸

Diagnosis of Fetal Varicella Zoster Virus Infection

- *Features of fetal varicella syndromes:* Polyhydramnios, limb defects, skin scarring, encephalitis, cortical atrophy, cortical cataract, optic atrophy, developmental delay, fetal growth restriction.
- Amniotic fluid PCR confirms fetal infection by detection of VZV DNA, but not necessarily congenital varicella syndrome.

Management

- If maternal infection has occurred prior to 20 weeks of gestation, serial ultrasound is recommended from 5 weeks since initial infection or from 16 weeks, whichever is earliest. Termination of pregnancy is advisable if congenital varicella syndrome is diagnosed after maternal infection, which has occurred before 20 weeks.²¹
- Varicella zoster immunoglobulin should be given in a nonimmune female following exposure to varicella zoster virus within 10 days.²¹
- Oral acyclovir can be given as part of postexposure prophylaxis within 24 hours of the development of rash, as it decreases its severity and improvement in constitutional symptoms.²¹

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